

Teorema de derivación bajo el signo integral

Teorema 1. Sea (X, Σ, μ) un espacio de medida y $f: X \times [a, b] \rightarrow \mathbb{K} = \mathbb{R} \text{ o } \mathbb{C}$ tal que $-\infty \leq a < b \leq \infty$. Definimos $F: [a, b] \rightarrow \mathbb{K}$ mediante

$$\forall t \in [a, b] : F(t) := \int_X f(x, t) \, d\mu(x).$$

Supongamos que $\exists \frac{\partial f}{\partial t}(x, t)$ para todo $(x, t) \in X \times [a, b]$ y $|\frac{\partial f}{\partial t}(x, t)| \leq g(x)$ para $g \in \mathcal{L}^1(X, \mu)$.

$$\implies F \text{ es diferenciable en } [a, b] \text{ y } \forall t \in [a, b] : F'(t) = \int_X \frac{\partial f}{\partial t}(x, t) \, d\mu(x).$$

Demostración: Definimos $\forall h \neq 0 : F_h(x, t) = \frac{f(x, t+h) - f(x, t)}{h}$. Tenemos que

$$F'(t) = \lim_{h \rightarrow 0} \frac{F(t+h) - F(t)}{h} = \lim_{h \rightarrow 0} \int_X \frac{f(x, t+h) - f(x, t)}{h} \, d\mu(x) = \lim_{h \rightarrow 0} \int_X F_h(x, t) \, d\mu(x).$$

Como $\forall (x, t) \in X \times [a, b] : \exists \frac{\partial f}{\partial t}(x, t)$, se tiene que $\lim_{h \rightarrow 0} F_h(x, t) = \frac{\partial f}{\partial t}(x, t)$. Además, por el teorema del valor medio en la variable t , para cada $(x, t) \in X \times [a, b]$ y $h \neq 0$, existe ξ entre t y $t+h$ tal que

$$|F_h(x, t)| = \left| \frac{f(x, t+h) - f(x, t)}{h} \right| = \left| \frac{\partial f}{\partial t}(x, \xi) \right| \leq g(x).$$

Por tanto, podemos aplicar el teorema de la convergencia dominada:

$$\forall t \in [a, b] : F'(t) = \int_X \lim_{h \rightarrow 0} F_h(x, t) \, d\mu(x) = \int_X \frac{\partial f}{\partial t}(x, t) \, d\mu(x). \quad \blacksquare$$

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